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Non-rectangular transducer arrays and associated devices, systems, and methods

Abstract

The present disclosure advantageously describes ultrasound imaging arrays that comprise ergonomic, non-rectangular shapes, as well as associated systems and methods. Non-rectangular transducer arrays allow for ergonomic probe shapes that improve patient comfort, maneuverability of the ultrasound device, and operator workflow. For example, an ultrasound imaging device can include an array of acoustic elements comprising a non-rectangular perimeter. The array includes a plurality of active elements configured to emit ultrasound energy and receive echoes corresponding to the emitted ultrasound energy, and a plurality of buffer elements surrounding the plurality of active elements at the non-rectangular perimeter of the array of acoustic elements. An edge seal comprising a sealing material is positioned at least partially around the plurality of buffer elements, and a buffer element of the plurality of buffer elements is spaced from at least one other buffer element by the sealing material of the edge seal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/271,115, filed on Feb. 24, 2021, which is a U.S. National Phase application under 35 U.S.C. § 371 of International Application No. PCT/EP2019/073255, filed on Aug. 30, 2019, which claims the benefit of U.S. Provisional Application No. 62/725,785, filed on Aug. 31, 2018. These applications are hereby incorporated by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates generally to ultrasound imaging and, in particular, to techniques for fabricating imaging components including a transducer array with non-rectangular shapes, and associated devices and systems.

BACKGROUND

(2) Ultrasound imaging is frequently used to obtain images of internal anatomical structures of a patient. Ultrasound systems typically comprise an ultrasound transducer probe that includes a transducer array coupled to a probe housing. The transducer array is activated to vibrate at ultrasonic frequencies to transmit ultrasonic energy into the patient's anatomy, and then receive ultrasonic echoes reflected or backscattered by the patient's anatomy to create an image. Such transducer arrays may include various layers, including some with piezoelectric materials, which vibrate in response to an applied voltage to produce the desired pressure waves. These transducers may be used to transmit and receive ultrasonic pressure waves through the various tissues of the body. The various ultrasonic responses may be further processed by an ultrasonic imaging system to display the various structures and tissues of the body.

(3) A transducer array typically includes a rectangular one-dimensional or two-dimensional matrix array of acoustic elements. In some aspects, rectangular ultrasound transducer arrays can pose challenges for an ultrasound technician. For example, in cardiac imaging (e.g., echocardiography), an external ultrasound probe may be positioned and precisely aligned between a patient's ribs to obtain images of the patient's heart. This can be difficult to do with an ultrasound probe that comprises a rectangular array of acoustic elements, because one or more corners of the array may restrict movement and alignment of the ultrasound probe between the patient's ribs and/or cause discomfort for the patient during the imaging procedure.

SUMMARY

(4) The present disclosure advantageously describes ultrasound imaging arrays that comprise ergonomic, non-rectangular shapes, as well as associated systems and methods. In one aspect, a non-rectangular array, which can also be referred to as a non-perpendicular array, can include one or more sealing materials around a perimeter of the array that forms an edge seal to provide structural integrity to less stable acoustic elements at or near the perimeter of the array. For example, a sealing material can be applied before, during, or after forming kerfs in an acoustic stack, where the kerfs are formed to divide the acoustic stack into individual acoustic elements. The sealing material can support and/or strengthen vulnerable areas of the acoustic stack while the kerfs are formed. In some aspects, non-rectangular transducer arrays allow for ergonomic probe shapes that improve patient comfort, maneuverability of the ultrasound device, and operator workflow.

(5) In one aspect, an ultrasound imaging device includes an array of acoustic elements comprising a non-rectangular perimeter. The array of acoustic elements further includes a plurality of active elements configured to emit ultrasound energy and receive echoes corresponding to the emitted ultrasound energy, and a plurality of buffer elements surrounding the plurality of active elements at the non-rectangular perimeter of the array of acoustic elements. The device further includes an edge seal comprising a sealing material positioned at least partially around the plurality of buffer elements, wherein a buffer element of the plurality of buffer elements is spaced from at least one other buffer element by the sealing material of the edge seal.

(6) In some embodiments, the non-rectangular perimeter comprises a curved segment. In some embodiments, the non-rectangular perimeter comprises a polygon. In some embodiments, each buffer element of a first portion of the plurality of buffer elements comprises a non-rectangular profile, the first portion of the plurality of buffer elements at an outer edge of the array of acoustic elements. According to some aspects, each buffer element of a second portion of the plurality of buffer elements comprises a rectangular profile, and the second portion of the plurality of buffer elements is spaced from the outer edge of the array of acoustic elements. According to other

aspects, the edge seal includes a first sealing material in direct contact with the non-rectangular perimeter of the array of acoustic elements, the first sealing material comprising a plurality of kerfs, and a second sealing material positioned around the first sealing material and disposed within the plurality of kerfs of the first sealing material. In some embodiments, the device further includes a processor chip coupled to a surface of the array of acoustic elements, wherein the processing chip comprises a non-perpendicular perimeter that aligns with the non-perpendicular perimeter of the array of acoustic elements. In another embodiment, the device further includes a housing, and the array of acoustic elements is coupled to the housing.

(7) According to another aspect of the present disclosure, a method for manufacturing an ultrasound imaging device includes removing material from a perimeter of an acoustic stack such that the acoustic stack comprises a non-rectangular perimeter, forming a first plurality of kerfs in the acoustic stack in a first direction, depositing a first sealing material at the non-rectangular perimeter of the acoustic stack, allowing the first sealing material to enter the first plurality of kerfs; curing the first sealing material, and forming a second plurality of kerfs in the acoustic stack and the first sealing material in a second direction to form an array of acoustic elements.

(8) In some embodiments, removing material from the perimeter of the acoustic stack comprises forming a curved segment. In other embodiments, removing material from the perimeter of the acoustic stack comprises forming a polygonal segment. In some aspects, allowing the first sealing material to enter the first plurality of kerfs comprises waiting for the first sealing material to advance between about 150 microns and about 250 microns into the acoustic stack. In another aspect, the first sealing material comprises an epoxy, and curing the first sealing material comprises directing ultraviolet light to the epoxy. In some embodiments, the method further includes depositing a second sealing material around the first sealing material, allowing the second sealing material to enter the second plurality of kerfs, and curing the second sealing material.

(9) In some embodiments, the method further includes removing material from a processing chip such that the processing chip comprises a non-perpendicular perimeter, and coupling the processing chip to the acoustic stack such that the non-perpendicular perimeter of the processing chip aligns with the non-perpendicular perimeter of the acoustic stack. In still other embodiments, the method further includes coupling the array of acoustic elements to a housing.

(10) Additional aspects, features, and advantages of the present disclosure will become apparent from the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Illustrative embodiments of the present disclosure will be described with reference to the accompanying drawings, of which:

(2) FIG. 1 is a schematic diagram of an ultrasound imaging system according to embodiments of the present disclosure.

(3) FIG. 2 is a cross sectional view of individual elements of a transducer array formed of an acoustic stack according to embodiments of the present disclosure.

(4) FIG. 3 is a flow diagram of a method for manufacturing a transducer array according to embodiments of the present disclosure.

(5) FIG. 4 is a top view of an array structure coupled to an integrated circuit (IC) layer in a stage of manufacturing according to embodiments of the present disclosure.

(6) FIG. 5 is a cross-sectional view of an array structure coupled to an IC layer in a stage of manufacturing according to embodiments of the present disclosure.

(7) FIG. 6 is a top view of an array structure coupled to an IC layer in a stage of manufacturing according to embodiments of the present disclosure.

- (8) FIG. 7 is a top view of an array structure coupled to an IC layer in a stage of manufacturing according to embodiments of the present disclosure.
- (9) FIG. 8 is a top view of an array structure coupled to an IC layer in a stage of manufacturing according to embodiments of the present disclosure.
- (10) FIG. 9 is a top view of an array structure coupled to an IC layer in a stage of manufacturing according to embodiments of the present disclosure.
- (11) FIG. 10 is a cross-sectional view of an array structure coupled to an IC layer in a stage of manufacturing according to embodiments of the present disclosure.
- (12) FIG. 11 is a flow diagram of a method for manufacturing a transducer array according to embodiments of the present disclosure.
- (13) FIG. 12 is a top view of an acoustic stack in a stage of manufacturing according to embodiments of the present disclosure.
- (14) FIG. 13 is a top view of an acoustic stack in a stage of manufacturing according to embodiments of the present disclosure.
- (15) FIG. 14 is a top view of an acoustic stack having a plurality of kerfs in a stage of manufacturing according to embodiments of the present disclosure.
- (16) FIG. 15 is a top view of an acoustic stack having a plurality of kerfs and a sealing material at a perimeter in a stage of manufacturing according to embodiments of the present disclosure.
- (17) FIG. 16 is a top view of an array structure formed of an acoustic stack in a stage of manufacturing according to embodiments of the present disclosure.
- (18) FIG. 17 is a top view of an array structure formed of an acoustic stack in a stage of manufacturing according to embodiments of the present disclosure.
- (19) FIG. 18 is a top view of a non-rectangular acoustic stack according to embodiments of the present disclosure.
- (20) FIG. 19 is a top view of a non-rectangular acoustic stack according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(21) For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, and specific language will be used to describe the same. It is nevertheless understood that no limitation to the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, and methods, and any further application of the principles of the present disclosure are fully contemplated and included within the present disclosure as would normally occur to one skilled in the art to which the disclosure relates. For example, while the ultrasound devices are described in terms of external imaging probes, it is understood that it is not intended to be limited to this application. The system is equally well suited to any application requiring imaging of an anatomy. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one embodiment may be combined with the features, components, and/or steps described with respect to other embodiments of the present disclosure. For the sake of brevity, however, the numerous iterations of these combinations will not be described separately.

(22) In FIG. 1, an ultrasound system 100 according to embodiments of the present disclosure is shown in block diagram form. An ultrasound probe 10 has a transducer array 12 comprising a plurality of ultrasound transducer elements or acoustic elements. In some instances, the array 12 may include any number of acoustic elements. For example, the array 12 can include between 1 acoustic element and 1000 acoustic elements, including values such as 2 acoustic elements, 4 acoustic elements, 36 acoustic elements, 64 acoustic elements, 128 acoustic elements, 300 acoustic elements, 812 acoustic elements, and/or other values both larger and smaller. In some instances, the acoustic elements of the array 12 may be arranged in any suitable configuration, such as a linear array, a planar array, a curved array, a curvilinear array, a circumferential array, an annular array, a phased array, a matrix array, a one-dimensional (1D) array, a 1.x dimensional array (e.g., a 1.5D

array), or a two-dimensional (2D) array. The array of acoustic elements (e.g., one or more rows, one or more columns, and/or one or more orientations) can be uniformly or independently controlled and activated. The array **12** can be configured to obtain one-dimensional, two-dimensional, and/or three-dimensional images of patient anatomy.

(23) While the present disclosure refers to external ultrasound imaging using an external ultrasound probe configured for imaging while positioned adjacent to and/or in contact with the patient's skin, it is understood that one or more aspects of the present disclosure can be implemented in any suitable array-based ultrasound imaging system. For example, aspects of the present disclosure can be implemented in intraluminal ultrasound imaging systems using an intracardiac (ICE) echocardiography catheter and/or a transesophageal echocardiography (TEE) probe, intravascular ultrasound (IVUS) imaging catheters, and/or transthoracic echocardiography (TTE) imaging device in some embodiments.

(24) Referring again to FIG. **1**, the acoustic elements of the array **12** may comprise piezoelectric/piezoresistive elements, lead zirconate titanate (PZT), piezoelectric micromachined ultrasound transducer (PMUT) elements, capacitive micromachined ultrasound transducer (CMUT) elements, and/or any other suitable type of acoustic elements. The acoustic elements of the array **12** are in communication with (e.g., electrically coupled to) electronic circuitry **14**. In some embodiments, such as the embodiment of FIG. **1**, the electronic circuitry **14** can comprise a microbeamformer (μ BF). In other embodiments, the electronic circuitry comprises a multiplexer circuit (MUX). The electronic circuitry **14** is located in the probe **10** and communicatively coupled to the transducer array **12**. In some embodiments, one or more components of the electronic circuitry **14** can be positioned in the probe **10**. In some embodiments, one or more components of the electronic circuitry **14**, can be positioned in a processor **28**, or processing system. In some aspects, some components of the electronic circuitry **14** are positioned in the probe **10** and other components of the electronic circuitry **14** are positioned in the processor **28**. The electronic circuitry **14** may comprise one or more electrical switches, transistors, programmable logic devices, or other electronic components configured to combine and/or continuously switch between a plurality of inputs to transmit signals from each of the plurality of inputs across one or more common communication channels. The electronic circuitry **14** may be coupled to elements of the array **12** by a plurality of communication channels. The electronic circuitry **14** is coupled to a cable **16**, which transmits signals including ultrasound imaging data to the processor **28**.

(25) In the processor **28**, the signals are digitized and coupled to channels of a system beamformer **22**, which appropriately delays each signal. The delayed signals are then combined to form a coherent steered and focused receive beam. System beamformers may comprise electronic hardware components, hardware controlled by software, or a microprocessor executing beamforming algorithms. In that regard, the beamformer **22** may be referenced as electronic circuitry. In some embodiments, the beamformer **22** can be a system beamformer, such as the system beamformer **22** of FIG. **1**, or it may be a beamformer implemented by circuitry within the ultrasound probe **10**. In some embodiments, the system beamformer **22** works in conjunction with a microbeamformer (e.g., electronic circuitry **14**) disposed within the probe **10**. The beamformer **22** can be an analog beamformer in some embodiments, or a digital beamformer in some embodiments. In the case of a digital beamformer, the system includes A/D converters which convert analog signals from the array **12** into sampled digital echo data. The beamformer **22** generally will include one or more microprocessors, shift registers, and or digital or analog memories to process the echo data into coherent echo signal data. Delays are effected by various means such as by the time of sampling of received signals, the write/read interval of data temporarily stored in memory, or by the length or clock rate of a shift register as described in U.S. Pat. No. 4,173,007 to McKeighen et al., the entirety of which is hereby incorporated by reference herein. Additionally, in some embodiments, the beamformer can apply appropriate weight to each of the signals generated by the array **12**. The beamformed signals from the image field are

processed by a signal and image processor **24** to produce 2D or 3D images for display on an image display **30**. The signal and image processor **24** may comprise electronic hardware components, hardware controlled by software, or a microprocessor executing image processing algorithms. It generally will also include specialized hardware or software which processes received echo data into image data for images of a desired display format such as a scan converter. In some embodiments, beamforming functions can be divided between different beamforming components. For example, in some embodiments, the system **100** can include a microbeamformer located within the probe **10** and in communication with the system beamformer **22**. The microbeamformer may perform preliminary beamforming and/or signal processing that can reduce the number of communication channels required to transmit the receive signals to the processor **28**.

(26) Control of ultrasound system parameters such as scanning mode (e.g., B-mode, M-mode), probe selection, beam steering and focusing, and signal and image processing is done under control of a system controller **26** which is coupled to various modules of the system **100**. The system controller **26** may be formed by application specific integrated circuits (ASICs) or microprocessor circuitry and software data storage devices such as RAMs, ROMs, or disk drives. In the case of the probe **10**, some of this control information may be provided to the electronic circuitry **14** from the processor **28** over the cable **16**, conditioning the electronic circuitry **14** for operation of the array as required for the particular scanning procedure. The user inputs these operating parameters by means of a user interface device **20**.

(27) In some embodiments, the image processor **24** is configured to generate images of different modes to be further analyzed or output to the display **30**. For example, in some embodiments, the image processor can be configured to compile a B-mode image, such as a live B-mode image, of an anatomy of the patient. In other embodiments, the image processor **24** is configured to generate or compile an M-mode image. An M-mode image can be described as an image showing temporal changes in the imaged anatomy along a single scan line.

(28) It will be understood that the processor **28** can comprise hardware, such as a computer processor, application-specific integrated circuit (ASIC), field-programmable gate array (FPGA), capacitors, resistors, and/or other electronic devices, software, or a combination of hardware and software. In some embodiments, the processor **28** is a single computing device. In other embodiments, the processor **28** comprises separate computer devices in communication with one another.

(29) FIG. 2 is an exemplary illustration of a portion of an acoustic element array **200**, according to aspects of the present disclosure. In some embodiments, the array **200** may be implemented as the transducer in the ultrasound imaging device. For example, in the illustrated embodiment of FIG. 2, the array **200** includes three acoustic elements **241**. In general, the array **200** can include any suitable number of acoustic elements **241**. The acoustic element **241** can be formed of a plurality of material layers (e.g., layers **240**, **242**, **244**, **246**, and/or other suitable layers). In some instances, the acoustic element **241** can be referenced as a pillar, such as a pillar including multiple material layers. The acoustic element array **200** may also include an acoustic matching layer **210**, a ground plane **221**, and a substrate **254**.

(30) Additional processes, steps, and features relating to forming acoustic stacks and acoustic arrays are described in U.S. Provisional App. No. 62/641,582, filed Mar. 12, 2018, the entirety of which is hereby incorporated by reference.

(31) In some embodiments, the substrate **254** is a semiconductor substrate that may form the base of the acoustic element array **200**. The substrate **254** may include materials such as silicon, silicon dioxide, aluminum oxide, germanium, and/or other suitable materials. In some embodiments, the substrate **254** is a flexible substrate, such as a polymer substrate, a polyimide substrate, e.g., Kapton®, and/or other suitable material. The material of the substrate **254** can be selected based on the application in some instances. For example, the flexible substrate can be used for a one-dimensional array. The flexible substrate can also be used for a two-dimensional or matrix array in

some instances. A semiconductor substrate, with, e.g., electrical interconnects contained therein, can be implemented in a two-dimensional or matrix array configured for three-dimensional imaging.

(32) In some embodiments, the acoustic matching layer **210** includes a pliable film, such as a polyurethane film. The matching layer **210** may be cast on the ground plane **221**. A bottom surface of the matching layer **210** may be in physical contact with the top surface of the ground plane **221** without an adhesive layer in between the matching layer and the ground plane **221**. In some embodiments, the matching layer **210** has a thickness of about 10 μm -200 μm and/or other suitable values, both larger and smaller. In some embodiments, the thickness of the matching layer **210** can be selected based on the acoustic center frequency of the transducer **220** (FIG. 2) or the array **200** (FIG. 3). For example, low frequency transducer designs (e.g., center frequency of about 0.5 MHz-5 MHz) can have about 100 μm -200 μm matching layer thickness, mid-range frequency transducer designs (e.g., center frequency of about 5 MHz-10 MHz) can have about 50 μm -150 μm matching layer thickness, and high frequency transducer designs (e.g., center frequency >10 MHz) can have about 10 μm -100 μm matching layer thickness. In some embodiments, the combination of the matching layer **210** and the ground plane **221** may be referred to as a metalized matching layer. The matching layer **210** can be or be part of an acoustic lens of the ultrasound imaging device. In some instances, the array **200** includes multiple matching layers **210**, such as different matching layers with different acoustic properties (e.g., acoustic impedance).

(33) In some embodiments, the ground plane **221** includes an upper layer **220** and a lower layer **322**. The upper layer **220** may include one or more polymers, such as polyether, polyester, or polyimide. In some embodiments, the upper layer **220** has a thickness of about 3-12 μm . In other embodiments, the upper layer **220** has a thickness of about 1-10 μm , 2-5 μm , 5-10 μm , and/or other suitable values, both larger and smaller. The lower layer **222** may include an electrode formed from a metal such as gold, silver, copper, aluminum, platinum, and/or other suitable materials. In some embodiments, the lower layer **222** has a thickness of about 3000 Å. In other embodiments, the lower layer **222** has a thickness of about 1000 Å-9000 Å, 2000 Å-3000 Å μm , or 4000 Å-6000 Å, and/or other suitable values, both larger and smaller. The electrode layer **222** can be configured to carry electrical current. In some embodiments, the thickness of the electrode layer **222** can be selected based on the amount of current being transmitted. In some instances, the thickness of the electrode layer **222** can be the same while the thickness of other layers (e.g., matching layer **210**, upper layer **220**) changes in different ultrasound devices. For example, the thickness of the electrode layer **222** can be the same while the thickness of the matching layer **210** changes according to the center frequency in different ultrasound devices. In some embodiments, the bottom surface of the matching layer **210** is in direct physical contact with the top surface of the upper layer **220** of the ground plane **221**. In some embodiments, the matching layer **210** is cast directly on the ground plane **221**. For example, the matching layer **210** may be cast on the ground plane **221** in a solvent casting process by depositing a liquid mixture with solvent and dissolved material on the ground plane **221** and then removing the solvent. Casting techniques such as the “doctor blade” technique may be used to cast the matching layer **210** on the ground plane. Other casting techniques for depositing the matching layer **210** on the ground plane **221** are also contemplated, including spin coating, drop casting, sputtering, printing, spray coating, blade coating, solution shearing, and other techniques.

(34) The attachment of the matching layer **210** to the ground plane **221** without adhesive may offer improved imaging performance compared to existing methods for forming transducers. Furthermore, the method of casting the matching layer **210** directly on the ground plane **221** offers manufacturing benefits. For example, the matching layer **210** is cast on the ground plane **221** before attaching the ground plane to the one or more acoustic elements **241**. Since layers within the one or more acoustic elements **241** may be sensitive to high temperature and pressures, combining the matching layer **210** with the ground plane **221** separately may avoid damage to the one or more

acoustic elements **241**. Furthermore, casting the matching layer **210** directly on the ground plane **221** may result in flatter layers than existing methods because the matching layer **210** and ground plane **221** are not offset by imperfections in the layers within the one or more acoustic elements **241** during manufacturing. The improved flatness may offer better imaging performance of the completed ultrasound transducers. Additionally, the matching layer **210** and ground plane **221** may be prefabricated on a large scale which may lower manufacturing costs and allow inspection of the matching layer **210** and ground plane **221** before integration with more expensive components with the one or more acoustic elements **241**. The ability to inspect earlier in the process may help to avoid unnecessary waste of expensive components.

(35) The ground plane **221** may be attached to the one or more acoustic elements **241**. In some embodiments, the ground plane **221** is configured to provide an electrical ground path for the layers of the acoustic elements **241**. Each of the one or more acoustic elements **241** may include any combination of layers, such as a second matching layer **240**, piezoelectric element **242**, third matching layer **244** (or de-matching layer), and bump **246** (which may include graphite and/or other suitable conductive material). The acoustic element **241** may also include one or more of an underfill **248** and bond pads **252** (which may include gold and/or other suitable conductive material). In some embodiments, the ground plane **221** and matching layer **210** wrap around a portion of the layers of the one or more acoustic elements **241**. For example, the ground plane **221** and matching layer **210** may wrap around the edges of the one or more acoustic elements **241**, such that the ground plane **221** is in contact with an outer sidewall of the one or more acoustic elements **241**.

(36) During manufacturing, an acoustic stack including multiple layers (layers **240**, **242**, **244**, **246** and/or other suitable layers) may be formed on the substrate **254**. The acoustic stack is then diced, which forms the individual acoustic elements **241** spaced from one another by gaps or kerfs **250**. In the embodiment of FIG. 2, the kerfs **250** are air-filled kerfs, or more simply, air kerfs. However, in other embodiments, the kerfs **250** can include any suitable type of kerf, such as a material-filled kerf. In some embodiments, the ground plane **221** is attached to the one or more acoustic elements **241** with an adhesive layer **230** disposed between the bottom surface of the ground plane and the top surface of the one or more acoustic elements **241**. In particular, the lower layer **322** of the ground plane **221** may be attached to the one or more acoustic elements **241**. The ground plane **221** and matching layer **210** may be disposed over multiple acoustic elements **241** and over the kerfs **250** between the acoustic elements **241**.

(37) Dividing the acoustic stack **200** into individual acoustic elements by forming kerfs is further explained below with respect to FIGS. 3-10. The acoustic array resulting from the kerfs is typically coupled to and encased in a housing filled with an encapsulating material. The encapsulating material can easily infiltrate into the air kerfs between the acoustic elements causing the air kerfs to be completely filled or partially filled instead of non-filled. One approach to protecting the air kerfs is to wrap all surfaces or sides of the array with a sealing film. However, the sealing film increases the footprint of the imaging component, which may not be desirable since catheters are space-limited. In addition, the wrapping of the ground plane may not completely seal the sides or surfaces of the array structure from infiltration of cleaning fluids, epoxies, or window material that are applied in subsequent fabrication process steps. In that regard, in some aspects, it can be beneficial to apply an edge seal around an imaging array that comprises air kerfs in order to seal the array.

(38) A method **300** of manufacturing a transducer array is described with reference made to FIGS. 3-10. FIG. 3 is a flow diagram of a method **300** of manufacturing a transducer array, or imaging component, according to embodiments of the present disclosure. It is understood that additional steps can be provided before, during, and after the steps of method **300**, and some of the steps described can be replaced or eliminated for other embodiments of the method. The steps of the method **300** can be carried out by a manufacturer of a catheter. FIG. 4 is a top view of an array structure **600** coupled to the IC layer **326** in a stage of manufacturing according to embodiments of

the present disclosure. FIG. 5 is a cross-sectional view of the array structure **600** coupled to the IC layer **326** in a stage of manufacturing according to embodiments of the present disclosure. FIG. 6 is a top view of the array structure **600** sealed with a sealing material **810** in a stage of manufacturing according to embodiments of the present disclosure. FIG. 7 is a top view of the array structure **600** under a wicking process in a stage of manufacturing according to embodiments of the present disclosure. FIG. 8 is a top view of the array structure **600** after the wicking process is completed in a stage of manufacturing according to embodiments of the present disclosure. FIG. 9 is a top view of the array structure **600** after excess sealing material is removed in a stage of manufacturing according to embodiments of the present disclosure. FIG. 10 is a cross-sectional view of the imaging component including the array structure **600** in a stage of manufacturing according to embodiments of the present disclosure.

(39) Additional processes, steps, and features relating to forming transducer arrays are described in U.S. Provisional App. No. 62/403,267, filed Oct. 3, 2016, and U.S. Provisional App. No. 62/434,568, filed Dec. 15, 2016, each of which is hereby incorporated by reference.

(40) Referring to the step **310** of the method **300** and FIGS. 4 and 5, in an embodiment, an array of acoustic elements **614** separated by air kerfs **618** is formed, for example, using a machining or dicing process or any suitable process. The acoustic elements **614** form part of an array structure **600** shown in in FIGS. 4 and 5.

(41) Referring to the step **320** of the method **300** and FIGS. 4 and 5, in an embodiment, a plurality of buffer elements **612** surrounding the array of acoustic elements **614** are formed. The plurality of buffer elements **612** is separated by gaps **616**. The buffer elements **612** do not include transducer functionalities. The buffer elements **612** do not emit ultrasound energy when activated. The buffer elements **612** can provide array uniformity and function as a buffer to protect the acoustic elements **614**, as described in greater detail herein. The buffer elements **612** form part of the array structure **600**.

(42) FIG. 4 illustrates a top view of the array structure **600** bonded to a top plane **650** of the IC layer **326**. The array structure **600** can be uniformly shaped and can have a rectangular shape or a square shape. As shown, the acoustic elements **614** are arranged in rows and columns. The buffer elements **612** are positioned on the outer-most rows and outer-most columns of the array structure **600** surrounding the acoustic elements **614**. The buffer elements **612** define the sides of the array structure **600**. The buffer elements **612** are separated from the acoustic elements **614** by air kerfs **620**. In some embodiments, the acoustic elements **614** and the buffer elements **612** can be uniformly spaced. Thus, the air kerfs **618** and **620** and the gaps **616** can have substantially similar widths and can be aligned to each other.

(43) Dimensions of the array structure **600** may vary in different embodiments. In some embodiments, the acoustic elements **614** can have lengths **602** between about 90 μm to about 130 μm and widths **604** between about 90 μm to about 130 μm . The widths **622** of the air kerfs **618**, the widths **624** of the gaps **616**, and the widths **626** of the air kerfs **620** can be between about 18 μm to about 30 μm . The buffer elements **612** can be sized to provide a buffering region with at least a depth **628** of about 100 μm for the array structure **600**.

(44) FIG. 5 illustrates a cross-sectional view of the array structure **600** coupled to the IC layer **326** taken along the line **601** of FIG. 4. The acoustic elements **614** and the buffer elements **612** include a matching layer **680a**, a piezoelectric layer **680b**, a dematching layer **680c**, and a bump layer **680d**. The matching layer **680a** matches the acoustic impedance of the piezoelectric layer **680b** to that of the body being diagnosed. The piezoelectric layer **680b** transmits ultrasound waves and receives echoes reflected off target tissue structures. The dematching layer **680c** reflects backward ultrasound waves travelling from the backside of the piezoelectric layer **680b**. The bump layer **680d** includes flip-chip bumps **691**. The matching layer **680a**, the dematching layer **680c**, and the bump layer **680d** can be composed of suitable conductive materials. The piezoelectric layer **680b** can be composed of lead zirconium titanate (PZT). The IC layer **326** includes bump pads **692**

coupled to the flip-chip bumps **691**. An underfill material **693** fills the region between the bump layer **680d** and the IC layer **326**. The outer edges of the array structure **600** can be plated with a metalized ground edge plating **710**, which may be composed of any suitable conductive material (e.g., gold).

(45) Referring to the step **330** of the method **300** and FIGS. **6** and **7**, in an embodiment, at least a portion of the gaps **616** between the plurality of buffer elements **612** are filled with a sealing material **810**. The ground edge plating **710** on top of the array structure **600** is not shown in FIG. **8** for clarity of illustration. For example, the sealing material **810** is applied around sides **630** of the array structure **600** and wicked into the gaps **616** between the plurality of buffer elements **612**. The sealing material **810** can be a curable ultraviolet (UV) epoxy material or any suitable material. FIG. **6** shows the array structure **600** with the sealing material **810** surrounding the sides **630**. FIG. **7** shows the sealing material **810** wicking or spreading into the gaps **616**. The spreading is shown by the arrows in FIG. **7**. The sealing material **810** is allowed to wick into a pre-determined portion of the gaps **616** (e.g., before reaching the air kerfs **618** and **620**) as shown by the dotted lines **901**. The pre-determined portion can vary in different embodiments. In some embodiments, the pre-determined portion includes a depth **930** of at least 20 μm from an outer-boundary of the buffer elements **612**. In some embodiments, the sealing material **810** can be cured by applying a UV activating light to the sealing material before the sealing material reaches the air kerfs **618**, **620**. In other embodiments, the sealing material **810** can be cured by any suitable process, such as by applying heat, chemicals, and/or wavelengths of light other than ultraviolet light (e.g., visible light, infrared light, etc.).

(46) Referring to the step **340** of the method **300** and FIG. **8**, in an embodiment, the sealing material **810** filling the gaps **616** between the plurality of buffer elements **612** is cured such that the array of acoustic elements **614** remain spaced apart by the air kerfs **618** and **620**. FIG. **8** shows that the sealing material **810** is spread into the gaps **616** reaching the pre-determined portion. The curing can include applying a UV light to stop the spreading or wicking of the sealing material **810** when the pre-determined portion is filled.

(47) Referring to the step **350** of the method **300** and FIGS. **8** and **9**, in an embodiment, excess sealing material **810** is removed, for example, by dicing along the dashed lines **1010** in FIG. **8**. FIG. **9** shows the array structure **600** after the excess sealing material **810** is removed. In some embodiments, the remaining sealing material **810** can have a thickness **812** between about 20 μm to about 40 μm .

(48) Referring to the step **360** of the method **300** and FIG. **10**, in an embodiment, a ground plane **1230** is bonded to the top of the array structure **600** to form the imaging component. FIG. **10** illustrates a cross-sectional view of the imaging component including the array structure **600** taken along the line **1101** of FIG. **9**. The ground plane **1230** can be a polyester film with gold metallization. Dimensions of the ground plane **1230** may vary in different embodiments. In some embodiments, the ground plane **1230** can have a thickness **1232** of about 5 μm . The ground plane **1230** provides an electrical ground return for the array structure **600**. For example, the outer-most bump pads **692** are connected to ground connections **694** as shown in FIG. **10**.

(49) After forming the array, the array can be coupled to a probe housing, such as a head of the probe housing. An encapsulating material can be applied to the head of the probe housing to secure the array within the probe housing. The encapsulating material may include polydimethylsiloxane (PDMS), polyurethane, UV adhesives, or any suitable material that have desirable characteristics such as acoustic properties, bonding strength, and ease to work with during manufacturing.

(50) The use of the sealing material **810** around the array structure **600** and partially filling the gaps **616** between the buffer elements **612** prevent the encapsulating material from wicking into the air kerfs **618** between the acoustic elements **614**. As described above, the sealing material **810** can have a thickness **812** between about 20 μm to about 40 μm . Thus, the disclosed embodiments can create air-filled kerfs with a minimal increase in the size of the imaging component. For example,

the disclosed embodiments can be applied to fabricate a transducer array for external ultrasound imaging, where a head of the ultrasound probe carrying the transducer array can be ergonomically positioned between ribs of the human body. In addition, the disclosed fabrication method is suitable for bulk production and automation.

(51) In some aspects, rectangular ultrasound transducer arrays can pose challenges for an ultrasound technician. For example, in cardiac imaging (e.g., echocardiography), an external ultrasound probe may be positioned and precisely aligned between a patient's ribs to obtain images of the patient's heart. This can be difficult to do for an ultrasound probe that comprises a rectangular array of acoustic elements, because one or more corners of the array may restrict movement and alignment of the ultrasound probe between the patient's ribs and/or induce pain or discomfort for the patient during the imaging procedure. Accordingly, it may be beneficial to produce ultrasound imaging arrays that comprise non-rectangular shapes, or non-perpendicular shapes, that are advantageously ergonomic. Non-rectangular transducer arrays can be coupled to, or placed within, ergonomic, non-rectangular housings that can improve patient comfort during imaging procedures, as well as operator workflows. For example, because a non-rectangular array and/or housing can be more easily maneuvered in restricted locations, such as between the patient's ribs, the operator can more easily orient the probe to achieve the desired imaging plane, and with reduced discomfort to the patient.

(52) However, producing two-dimensional arrays of acoustic elements that comprise non-rectangular shapes can be challenging. As will be explained further below, non-rectangular arrays will typically include some non-rectangular acoustic elements that are structurally unstable, and susceptible to breaking away from the acoustic stack. It is therefore desirable to employ manufacturing processes for producing non-rectangular arrays that avoid creating unstable acoustic elements that are likely to break off during assembly, or during an imaging procedure. FIGS. **11-17** illustrate a method for manufacturing an ergonomic, non-rectangular ultrasound transducer array, according to some aspects of the present disclosure. FIG. **11** is a flow diagram describing various steps of a method for manufacturing a non-rectangular transducer array, according to some aspects of the present disclosure. FIGS. **12-17** illustrate an acoustic stack **720** at various steps of the process detailed in FIG. **11**.

(53) Referring to FIGS. **11-13** a rectangular acoustic stack **720** is provided (FIG. **12**), and material is removed from a perimeter of the acoustic stack **720**, such as from a corner **722** of the acoustic stack **720** (FIG. **13**). It will be understood that, for illustrative purposes, FIG. **12** illustrates only a portion or area of the acoustic stack **720**, rather than the entirety of the acoustic stack **720**.

(54) The acoustic stack **720** comprises a corner **722**. As explained above, corners of rectangular arrays can present challenges in imaging procedures, particularly where a probe must be guided to and placed between bodily structures, such as the ribs. Accordingly, it may be beneficial to alter the shape, such as the perimeter, of the acoustic stack **720** to provide a more ergonomic design. As shown in FIGS. **12** and **13**, the acoustic stack **720** can be made more ergonomic by removing material from the corner area **722** to produce a non-rectangular shape or perimeter. In step **510**, material is removed from the corner area **722** to form a chamfered edge **724**. In other embodiments, material can be removed by produce one or more curved segments and/or one or more polygonal shapes. For example, the shape of the acoustic stack **720** can be modified to include one or more of a chamfer, a bevel, a fillet, a round edge, or any combination of suitable geometric features. The material of the stack can be removed by, for example, cutting, dicing, grinding, etching, and/or by any other suitable shaping process. In some embodiments, the acoustic stack **720** is pre-formed to have a non-polygonal shape including any of the non-polygonal features described above such that no material needs to be removed from the acoustic stack **720** to create a non-polygonal shape.

(55) FIGS. **18** and **19** depict a polygonal transducer array **820**, and a non-polygonal, ellipsoidal transducer array **920**, respectively. Any suitable polygonal shape, ellipsoidal shape, regular shape, irregular shape, symmetrical shape, and/or non-symmetrical shape for the transducer array is

contemplated. In that regard, it is understood that FIGS. **18** and **19** illustrate a profile or a top view of the transducer array, such as a length and a width. The transducer array is three-dimensional in that it also has a height or a depth (e.g., as shown in FIGS. **2**, **5**, and **10**). In that regard, the transducer array can be described as a geometric prism in some instances. One or both of the transducer arrays **820**, **920** can be formed according to the method **500** set forth in FIG. **11**. For example, the transducer arrays can be formed by removing material from the corners of a rectangular acoustic stack, or forming the acoustic stacks **820**, **920** to have non-rectangular shapes without the need to remove material.

(56) Referring again to FIGS. **11-13**, in some embodiments, the non-rectangular acoustic stack **720** can be coupled, attached, or otherwise joined to a processor chip, or a computer chip, such as an ASIC. In some aspects, the computer chip can be described as an integrated circuit (IC), or IC layer. The computer chip, or IC layer, can comprise a similar or identical shape as the non-rectangular acoustic stack **720**. In that regard, in some embodiments, step **510** comprises simultaneously removing material from the rectangular acoustic stack **720** and a rectangular computer chip, such that the shapes of the acoustic stack **720** and the computer chip align. This can be performed before or after attaching the computer chip to the acoustic stack **720**. In other embodiments, the acoustic stack **720** and computer chip are shaped separately. The computer chip can include integrated logics and/or circuitries formed from a semiconductor material, such as silicon. The integrated logics and/or circuitries are configured to multiplex control signals, for example, generated by the processor **28** shown in FIG. **1**, and transfer the control signals to corresponding acoustic elements of an array. The controls signals can control the emission of ultrasound pulses and/or the reception of echo signals. In the reverse direction, the integrated logics and/or circuitries are configured to receive ultrasound echo signals reflected by target tissue and received by the acoustic elements. The integrated logics and/or circuitries convert the ultrasound echo signals into electrical signals and transfer the electrical signals through an interposer and an electrical cable **16** to the processor **28** for processing and/or display. The integrated logics and/or circuitries can be further configured to perform signal conditioning before transferring the signals. Signal conditioning may include filtering, amplification, and beamforming. In some embodiments, the computer chip may have a longer length than the acoustic stack **720** for coupling to an interposer.

(57) In step **520**, and as shown in FIG. **14**, the acoustic stack **720** is diced, cut, ground, etched, or otherwise machined to form a first plurality of kerfs **726**, which may also be described as gaps or spaces. The kerfs **726** are parallel and are formed in a first direction (horizontal) to separate a plurality of acoustic stack segments **728**. Although explained in terms of dicing, any process described above can be used to create the kerfs **726**, such as cutting and/or etching. In some embodiments, photolithography techniques can be used to create the kerfs **726**. In the illustrated embodiment, the kerfs **726** comprise air kerfs, meaning kerfs or channels that separate individual acoustic stack segments **728** or acoustic elements by air. In other embodiments, the kerfs **726** can be filled with a material, such that individual acoustic stack segments **728** and/or acoustic elements are separated from one another by the material.

(58) Particularly when air kerfs are used, such as in the acoustic stack **720** of FIG. **14**, it may be beneficial to provide a seal for the air kerfs **726** so that a foreign material (e.g., ultrasound gel, water, cleaning solution) does not find its way into the kerfs **726** of the acoustic stack **720**. Accordingly, in step **530**, shown in FIG. **15**, an edge seal **730** comprising a first sealing material **732** is applied around a perimeter of the acoustic stack **720** to seal the kerfs **726**, such that the first sealing material **732** is in direct contact with the non-rectangular perimeter of the acoustic stack **720**. The first sealing material **732** includes a first plurality of penetrating sections **734** partially disposed within the kerfs **726**. In step **540**, the penetrating sections **734** are formed by allowing the first sealing material **732** to enter into the first plurality of kerfs **726** by a certain distance, and in step **550**, the first sealing material **732** is cured. For example, the first sealing material **732** may

comprise a curable adhesive, potting material, or other fluidic material that can flow between kerfs **726**. In one embodiment, the first sealing material **732** comprises an epoxy or curable adhesive that can be cured by, for example, ultraviolet (UV) light, heat, air exposure, or any other suitable method. In some aspects, the process for forming the penetrating sections **734** can include waiting a pre-determined amount of time and/or waiting for the first sealing material **732** to advance a pre-determined distance into the kerfs **726**, and activating a curing process (e.g., applying UV light) to cure and/or solidify the first sealing material **732**, such that the first sealing material **732** ceases to advance into the kerfs **726** within the acoustic stack **720**. For example, in some embodiments, the first sealing material may be allowed to advance, or wick, into the kerfs **726** for a period of about 1 second to about 30 seconds. The amount of time allowed for the first sealing material to wick into the acoustic stack **720** may depend on the viscosity of the pre-cured sealing material. In some embodiments, the first sealing material **732** may be allowed to advance into the acoustic stack **720** between about 150 μm and about 1 mm, including between about 250 μm and about 350 μm . In some embodiments, the sealing material **732** may be allowed to advance into the acoustic stack **720** by a distance corresponding to a width of each acoustic stack segment **728** and/or acoustic element, such as a width corresponding to one half of an acoustic element, one acoustic element, two acoustic elements, etc.

(59) With reference to FIGS. **11** and **15**, in step **560** a second plurality of kerfs **736** is formed in the acoustic stack **720** in a second direction (vertical). In that regard, in the embodiment, of FIG. **15**, the direction of the second plurality of kerfs **736** is perpendicular to the direction of the first plurality of kerfs **726**. Forming the second plurality of kerfs **736** divides the acoustic stack segments **728** shown in FIGS. **12** and **13** into a two dimensional array **750** of acoustic elements, wherein the array **750** comprises active elements **729** and buffer elements **738**. Buffer elements **738** may be described as acoustic elements of the array **750** that are inactive. For example, buffer elements **738** may be considered inactive due to mechanical restriction by sealing material **732** disposed between and/or around the buffer elements **738**, electrical connections or a lack thereof that prevent the buffer elements **738** from being driven to emit and receive ultrasound energy, or both. In other words, although the buffer elements **738** are formed from the same acoustic stack **720** as the active elements **729**, the buffer elements **738** are considered inactive or inert.

(60) The buffer elements **738** are positioned at or near the perimeter of the array **750**. A first portion **738a** of the buffer elements in the acoustic stack of FIG. **14** comprise non-rectangular shapes or profiles (e.g., triangular), and are located at or near the chamfered edge **724** of the array **750**. The chamfered edge **724** comprises a portion of the perimeter of the array **750**. A second portion **738b** of the buffer elements comprises rectangular shapes or profiles. One or more buffer elements of the second portion **738b** may be spaced from the outer perimeter of the array by one or more other buffer elements, while the buffer elements of the first portion **738a** are positioned at the perimeter of the array **750**, or directly adjacent the perimeter. In some aspects, the buffer elements **738** can be described as forming the perimeter or edge of the array **750**.

(61) As mentioned above, some of the buffer elements, particularly the first portion **738a** of the buffer elements may be unstable without the reinforcement of the first sealing material **732**. Thus, applying the first sealing material **732** before forming the second plurality of kerfs **736** can add support and reinforcement to the buffer elements **738** located at or near the outer perimeter or outer edge of the array **750**, particularly the buffer elements of the first portion **738a** that comprise non-rectangular cross sections and/or are more susceptible to breaking away from the array **750**.

(62) In step **570**, a second sealing material **740** is deposited over, or in addition to, the cured first sealing material **732**, such that the second sealing material **740** at least partially penetrates or enters the second plurality of kerfs **736** to seal the second plurality of kerfs **736** of the transducer array **750**. As with the first sealing material, in step **580**, the second sealing material **740** is allowed to advance within the second plurality of kerfs **736** to form a second plurality of penetrating portions **742**. The second sealing material **740** may comprise the same type of material as the first sealing

material **732**, or may comprise a different material. The second sealing material **740** may comprise a separate layer of material disposed over, or surrounding the first sealing material **732**. In other embodiments, the second sealing material **740** may not be applied and cured. For example, in some embodiments, a wrap is applied around the transducer array **750**, instead of the second sealing material **740**.

(63) In step **590**, excess sealing material from the first and/or second sealing materials **732**, **740** is trimmed, cut, diced, ground, or otherwise removed, leaving the first and second penetrating portions **734**, **742** in place to seal the array **750** and/or stabilize the buffer elements near the perimeter of the array **750**. The array **750** is then attached to a housing, for example within a head of the probe housing. The housing can be an external ultrasound probe, ICE catheter, TEE probe, IVUS catheter, TTE probe, or any array-based ultrasound device. As explained above, the array **750** can be attached to the probe housing using a filling adhesive and/or potting material. Because the array **750** comprises a non-rectangular shape, the array **750** may fit within a more ergonomically designed probe housing such that the probe can be more easily maneuvered and placed to obtain an ultrasound image. For example, the non-rectangular probe may be more easily maneuvered to obtain images between the ribs of a patient.

(64) It will be understood that other processing steps may be performed in addition to those described with respect to FIGS. **11-17**. For example, in one embodiment, a third sealing material may be applied in the first and second pluralities of kerfs **726**, **736** between the active acoustic **729** elements and/or the buffer elements **738**, displacing the air in the kerfs **726**, **736**. The third sealing material may comprise an acoustically compatible material configured to allow for activation and use of individual elements of the array **750**. In some embodiments, an acoustic matching layer and/or a ground plane (e.g., **210**, **220**, FIG. **2**) can be attached to top side of the array **750**. The acoustic matching layer and/or the ground plane can function with the edge seal **730** to seal the array **750**. It will also be understood that one or more steps, features, and/or components described with respect to FIGS. **2-10** can be used in conjunction with the steps, features, and/or components described with respect to FIGS. **11-17**.

(65) Persons skilled in the art will recognize that the apparatus, systems, and methods described above can be modified in various ways. Accordingly, persons of ordinary skill in the art will appreciate that the embodiments encompassed by the present disclosure are not limited to the particular exemplary embodiments described above. In that regard, although illustrative embodiments have been shown and described, a wide range of modification, change, and substitution is contemplated in the foregoing disclosure. It is understood that such variations may be made to the foregoing without departing from the scope of the present disclosure. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the present disclosure.

Claims

1. An ultrasound imaging device comprising: an array of acoustic elements comprising a non-rectangular perimeter, the array of acoustic elements including a plurality of active elements configured to emit ultrasound energy and receive echoes corresponding to the emitted ultrasound energy; a plurality of kerfs separating the acoustic elements of the array; and an edge seal comprising at least one sealing material applied at least partially around the array, wherein a plurality of sections of the at least one sealing material is partially disposed within the plurality of kerfs to seal the plurality of kerfs, wherein the at least one sealing material comprises a non-rectangular portion at the non-rectangular perimeter of the array, wherein the plurality of sections comprises a first section and a second section comprising different locations along the non-rectangular portion such that an end of the first section is further inward than an end of the second section.

2. The device of claim 1, wherein the non-rectangular perimeter comprises a curved segment.
 3. The device of claim 1, wherein the non-rectangular perimeter comprises a polygon.
 4. The device of claim 1, further comprising a plurality of buffer elements surrounding the plurality of active elements at the non-rectangular perimeter of the array of acoustic elements.
 5. The device of claim 4, wherein each buffer element of a first portion of the plurality of buffer elements comprises a non-rectangular profile, the first portion of the plurality of buffer elements at an outer edge of the array of acoustic elements.
 6. The device of claim 5, wherein each buffer element of a second portion of the plurality of buffer elements comprises a rectangular profile, the second portion of the plurality of buffer elements spaced from the outer edge of the array of acoustic elements.
 7. The device of claim 1, wherein the at least one sealing material comprises: a first sealing material in direct contact with the non-rectangular perimeter of the array of acoustic elements; and a second sealing material positioned around the first sealing material.
 8. The device of claim 7, wherein the plurality of sections comprises: a plurality of first sections of the first sealing material partially disposed within the plurality of kerfs; and a plurality of second sections of the second sealing material partially disposed within the plurality of kerfs.
 9. The device of claim 1, further comprising a processor chip coupled to a surface of the array of acoustic elements, wherein the processing chip comprises a non-rectangular perimeter that aligns with the non-rectangular perimeter of the array of acoustic elements.
 10. The device of claim 1, further comprising a housing, wherein the array of acoustic elements is coupled to the housing.
 11. The device of claim 1, wherein the plurality of sections is positioned into the plurality of kerfs by a predetermined distance.
 12. The device of claim 11, wherein the predetermined distance is same for the first section and the second section such that the end of the first section extending further inward than the end of the second section is attributable to the first section and the second section comprising the different locations along the non-rectangular portion of the at least one sealing material.
 13. The device of claim 1, wherein the plurality of kerfs comprise a first plurality of kerfs formed in a first direction and a second plurality of kerfs formed in a second direction perpendicular to the first direction.
 14. The device of claim 13, wherein the at least one sealing material comprises a first sealing material and a second sealing material, wherein the plurality of sections comprises: a plurality of first sections of the first sealing material partially disposed within the first plurality of kerfs; and a plurality of second sections of the second sealing material partially disposed within the second plurality of kerfs.
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